FULL STACK DEVELOPMENT

Introduction to Full Stack Web Applications

1. Introduction

Full Stack Development is the process of building both the **frontend** and **backend** of a web application and working with its **database**. A full stack developer understands how these parts connect and work together to provide a complete application to users.

Example: In an online shopping website, product images, prices, and buttons are part of the frontend. The backend processes actions such as logging in or placing an order. The database stores product, customer, and order information.

2. Main Components of Full Stack Development

Component	Purpose	Examples
Frontend	The part users see and interact with, including pages, forms, buttons, and menus.	HTML, CSS, JavaScript, React.js
Backend	Processes requests, applies application logic, and communicates with the database when required.	Python, Django, Django REST Framework
Database	Stores and organizes data so it can be saved, retrieved, and updated.	PostgreSQL, MySQL, SQLite

3. How the Components Work Together

A user interacts with the frontend, which sends a request to the backend. The backend processes the request and may read or update information in the database. It then sends a response to the frontend, and the browser displays the result. This communication is commonly handled using HTTP or HTTPS.

Example: When a student clicks “View Result” on a college website, the frontend sends a request. The backend retrieves the result from the database and returns it. The browser displays the result to the student.

4. Benefits of Full Stack Development

- **Complete understanding:** Helps developers understand how different application components connect.
- **Efficient development:** Makes it easier to coordinate work across frontend and backend components.
- **Problem solving:** Helps identify and troubleshoot issues across the application.
- **Versatile skills:** Knowledge of multiple layers is useful when creating and maintaining web applications.

PYTHON FULL STACK DEVELOPMENT

Technologies, Workflow, and Conclusion

5. Technologies Used in a Python Full Stack Application

Technology	Role	Description
HTML	Frontend	Defines the structure and content of a web page.
CSS	Frontend	Controls layout, colours, spacing, and fonts.
JavaScript	Frontend	Adds interactive behaviour to the page.
React.js	Frontend	Builds user interfaces using reusable components.
Python	Backend	Implements the application's logic.
Django	Backend	A Python framework for building web applications.
Django REST Framework	API layer	Helps create REST APIs for frontend-backend communication.
SQL	Database language	Used to retrieve, insert, update, and delete relational data.
PostgreSQL / MySQL / SQLite	Database	Examples of relational database management systems.

6. Typical Workflow of a Web Application

Step 1: The user opens a website and performs an action.
Step 2: The frontend sends a request to the backend.
Step 3: The backend validates and processes the request.
Step 4: The backend reads or updates the database if needed.
Step 5: The backend sends a response to the frontend.
Step 6: The frontend displays the result to the user.

7. Conclusion

Full Stack Development combines the frontend, backend, and database to create a complete web application. HTML, CSS, JavaScript, and React.js can be used for the frontend. Python and Django can handle backend logic, while Django REST Framework can provide APIs. A relational database stores application data, which the backend accesses as required. Understanding these technologies and how they work together is a foundation for developing useful and reliable web applications.